

BudgetRAG / MemAlign-Qwen: Resource-Aware Retrieval and Context Allocation for Grounded LLM Inference

Internship Progress Report

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Abstract

Long-context Retrieval-Augmented Generation (RAG) improves grounding by exposing the generator to retrieved evidence, but it also increases prompt length, latency, and estimated decoder KV-cache memory. This internship project studies resource-aware retrieval and context allocation: given a query and retrieved candidates, choose a retrieval-context action that preserves grounded answer quality while reducing token, latency, and estimated KV-cache costs. The work starts from a benchmark foundation with retriever registration, benchmark CLI, and retrieval metrics; then adds BudgetRAG context policies, deterministic adaptive profiles, generation feedback, RAGAS/MiMo answer labels, RLAIIF-style answer/context labeling, reward/preference construction, and offline selector baselines. The strongest current results are diagnostic rather than production claims: 192 MiMo answer labels, 192 merged MiMo context labels with 177 clean usable rows, context sufficiency rate 0.591, mean selected chunks 1.276 versus mean irrelevant chunks 3.708, and context-reward ablations that change 140 of 192 reward rows. Multi-seed held-out selector evaluation shows that shrinkage-smoothed best average is currently the strongest full-coverage non-oracle baseline, while learned selectors remain data-limited. The project does not claim human preference learning, online reinforcement learning, DPO/PPO/GRPO training, or runtime KV pruning. Future work should expand logged query groups, retrieval diversity, multi-judge audits, and local Qwen/KV profiling.

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1 Introduction

Retrieval-Augmented Generation (RAG) combines external retrieval with language-model generation so that answers can be grounded in retrieved evidence rather than only in parametric memory [8]. The practical baseline is often simple: retrieve the top-k documents and place most of their content into the prompt. That baseline is easy to implement, but it is not resource-aware. Retrieved context can contain irrelevant chunks, longer prompts increase latency and cost, and long sequence lengths increase decoder KV-cache memory for local LLM inference.

BudgetRAG / MemAlign-Qwen addresses this trade-off. The core question is: for a given query and a set of retrieved candidates, which retrieval strategy, context policy, and context budget should be selected to maximize grounded answer quality while controlling token count, latency, and estimated KV-cache footprint?

The MemAlign-Qwen motivation is local and privacy-constrained inference. In later phases, private documents should be processed by trusted local models rather than remote APIs. In the current phase, BudgetRAG reduces context before inference and uses RLAIIF-style context labels as future evidence for local Qwen profiling and possible KV evidence masks. The current report is careful about the boundary: the system is an offline research pipeline, not a deployed production policy.

2 Problem Formulation

For a query q and a candidate set C returned by one or more retrievers, the system chooses an action:

$$a = (\text{retrieval strategy, fusion strategy, context policy, budget, adaptive profile, generator model}).$$

The objective is an offline reward that trades off answer quality and evidence support against resource costs:

$$R = w_q Q + w_s S - w_t T - w_l L - w_k K - P_{\text{unsupported}} - P_{\text{error}}.$$

Here Q is answer quality/correctness, S is evidence or context support, T is normalized token cost, L is normalized latency, K is normalized estimated KV-cache cost, and the penalty terms capture unsupported claims and failed/invalid generations or labels.

The state includes query features, retrieval diagnostics, action metadata, and cost estimates. The evaluation is **offline logged-candidate evaluation**: a selector can only choose among actions that were already logged for a query. It is not online RL, and selector artifacts explicitly keep `runtime_default_replacement=false`.

3 Related Work

3.1 RAG and RAG Evaluation

RAG introduced a practical way to combine retrieval with neural generation for knowledge-intensive tasks [8]. BEIR showed that retrieval quality varies across heterogeneous domains and that no single retriever should be treated as universally best [14]. RAGAS adds automated RAG evaluation signals such as answer relevance and faithfulness [5]. In this project, BEIR-style metrics motivate retrieval benchmarking, while RAGAS is used as an initial proxy before richer MiMo answer labels are introduced.

3.2 Prompt and Context Compression

LLMLingua, LongLLMLingua, and Selective Context study prompt/context compression for efficient LLM inference [6, 7, 10]. BudgetRAG differs in formulation: context compression is not a standalone pre-processing step, but part of a retrieval-context action evaluated under answer quality, context sufficiency, token cost, latency, and estimated KV cost.

3.3 KV-Cache Compression and Pruning

H2O, StreamingLLM, KIVI, SnapKV, and PyramidKV reduce runtime memory or attention cost through cache selection, quantization, or pruning [16, 15, 12, 11, 2]. BudgetRAG currently operates before inference by selecting a smaller or cleaner context. Runtime KV pruning is future work; context-level RLAIIF labels may later provide evidence-token masks.

3.4 Graph Retrieval, RLAIIF, and Contextual Bandits

GraphRAG highlights graph-based retrieval for local-to-global query-focused summarization [4]. Hybrid retrieval and Reciprocal Rank Fusion (RRF) are relevant for combining retrieval strategies [3]. RLAIIF and preference learning use AI feedback to construct reward or preference signals [1]; DPO is a later preference-optimization method, not implemented here [13]. Because the BudgetRAG action is a one-step query-time decision, offline contextual bandit framing is more appropriate than PPO/GRPO in the current data-limited phase [9].

4 System Overview

The project pipeline is:

Query → retriever / graph / hybrid retrieval → BudgetRAG context policy →
generator → answer/context feedback → reward/preference rows → offline selector
evaluation

The main implementation components are:

- rag-bench: benchmark, BudgetRAG, and RLAIIF CLI.
- Retriever registry: BM25 plus scaffolded graph/hybrid strategies.
- Context policies: legacy, char-budget, per-doc-budget, score-density, sentence-trim, evidence-aware, and adaptive-heuristic.
- Adaptive profiles: conservative, balanced, and aggressive.
- RLAIIF artifacts: rlaif_actions.jsonl, rlaif_feedback.jsonl, rlaif_rewards.jsonl, and rlaif_preferences.jsonl.
- Labelers: rlaif-label-answers, rlaif-label-contexts, and rlaif-label-pairs.
- Selectors: fixed, cheapest, best average, family-smoothed, shrinkage-smoothed, linear reward model, smoothed linear selector, and oracle logged.

The design keeps raw experiment outputs under ignored benchmark_results/; committed reports are curated summaries. This prevents raw judge outputs, downloaded Kaggle working directories, or secrets from becoming part of the repository history.

5 Phase 1A: Benchmark Foundation

Phase 1A created the benchmark foundation: a retriever registry, benchmark CLI, retrieval metrics, and traceable output records. The metric layer includes hit@k, MRR, nDCG, latency, and context size/cost fields. This phase made later BudgetRAG experiments reproducible: fixed policies, adaptive profiles, and RLAIIF builders all consume the same output format.

The current report does not invent Phase 1A numbers beyond the available records. Quantitative reporting starts from Phase 1B/1C, where committed reports contain concrete matrix outputs.

6 Phase 1B: BudgetRAG Fixed Context Policies

Phase 1B introduced fixed context policies. `legacy` preserves the older behavior, `char-budget` clips total characters, `per-doc-budget` avoids one long document dominating the prompt, `score-density` favors high score per length, `sentence-trim` avoids cutting inside sentences, and `evidence-aware` keeps lexical/query-aware evidence spans. Importantly, `evidence-aware` is not an answer-aware entailment verifier.

The Phase 1B smoke run used 10 SciFact queries, BM25, top-k 3, skipped generation, and a Qwen2.5-14B KV profile. The 1000-character variants kept about 1000 characters with compression around 0.216 and analytical KV savings around 900 units in the report. Quality columns are blank because generation was skipped.

7 Phase 1C: Adaptive Profiles

Phase 1C replaced a purely fixed context budget with a deterministic adaptive heuristic. The selector uses retrieval diagnostics such as score gap, normalized entropy, confidence, and long-document dominance.

Phase 1C.1 expanded validation to 50 SciFact BM25 queries. The heuristic selected `evidence-aware` for 48 queries and `per-doc-budget` for 2, but still selected budget 4000 for all 50 queries. This showed stable metadata but an overly conservative policy.

Phase 1C.2 added conservative, balanced, and aggressive profiles. Balanced and aggressive profiles created more diverse budget choices. For example, with matrix budget 1000, conservative kept 4000 characters for all 50 queries, balanced averaged 2260 kept characters, and aggressive averaged 999.8 kept characters. This was still a deterministic heuristic baseline, not a learned selector.

8 Phase 1C.3: Generation Feedback Layer

Retrieval-only metrics cannot determine whether compressed context still supports a grounded answer. Phase 1C.3 added a generation feedback layer. RAGAS answer relevancy was first joined back to BudgetRAG `query_results.jsonl`; then MiMo answer labels were added for richer answer correctness, evidence support, faithfulness, and unsupported-claim penalties.

The Phase 1D selector-smoke input contained 192 joined action-query rows from 64 action files. Model coverage was MiMo 96 rows, Llama 3.1 8B 48 rows, and Qwen3 32B 48 rows. Context-policy coverage was adaptive-heuristic 96, evidence-aware 48, and legacy 48. Budget coverage included 4000:48, 8000:48, 1000:36, 2000:36, 16000:12, and 32000:12.

The full MiMo answer-label run produced 192 labels, 192 valid JSON rows, 0 invalid JSON rows, 25 ambiguous labels, 1 judge error, and 191 scored labels. The Pearson correlation between MiMo quality

and RAGAS answer relevancy was 0.277. This low correlation supports the interpretation that RAGAS answer relevancy and MiMo answer quality measure related but not identical signals.

9 Phase 1D: RLAIIF Reward and Preference Pipeline

Phase 1D converts BudgetRAG logs into normalized RLAIIF-style datasets:

Command	Role
<code>rlaif-build</code>	Convert BudgetRAG rows into normalized actions and feedback.
<code>rlaif-label-answers</code>	Label answer correctness, support, faithfulness, and unsupported claims.
<code>rlaif-label-context</code>	Label context sufficiency, useful chunks, redundant chunks, irrelevant chunks, and missing evidence.
<code>rlaif-reward</code>	Build scalar rewards and pairwise preferences with quality guardrails.
<code>rlaif-split</code>	Split by <code>benchmark + query_id</code> to avoid row-level leakage.
<code>rlaif-train/eval</code>	Train and evaluate offline selector baselines.
<code>rlaif-label-pairs</code>	Directly judge action pairs for reward calibration audits.

The reward builder preserves provenance and missing reasons. Missing, ambiguous, or invalid labels do not become score zero. Preferences are generated within comparable query groups, with guardrails that reject efficiency-only wins when answer quality drops too much.

RLAIIF in this report means AI-feedback labeling for reward/preference construction. It is not full RLHF and it is not human preference learning.

10 Main Results

10.1 Full MiMo Context Labels

The first full context-label pass merged 192 action rows with no missing, unknown, duplicate, or conflicting action ids. It produced 177 clean usable labels, 15 ambiguous labels, 0 invalid JSON labels, and 1 error. The sufficiency rate was 0.591, with 110 sufficient contexts and 76 insufficient contexts.

Metric	Value
Merged labels	192
Clean usable labels	177
Ambiguous labels	15
Sufficiency rate	0.591
Mean selected chunks	1.276
Mean redundant chunks	0.286
Mean irrelevant chunks	3.708
Mean context quality	0.602
Mean evidence support	0.556
Mean minimality	0.947

The main qualitative signal is that the judge often identifies only one or two useful chunks while marking several chunks as irrelevant. This supports context-level supervision rather than answer-only reward.

10.2 Context Reward Ablation

Context labels materially changed reward/preference construction. The answer-only baseline created 722 preferences. Context penalty 0.25 created 952 preferences and changed 140 reward rows; 108 changes were negative and 32 were positive. Stronger penalties were more aggressive and compressed the reward scale.

Setting	Preferences	Changed rows	Negative deltas	Mean changed delta
answer-only	722	—	—	—
context penalty 0.25	952	140	108	-0.212
context penalty 0.50	946	140	108	-0.316
context penalty 1.00	944	140	108	-0.525

Penalty 0.25 is the conservative non-default candidate. Penalty 1.00 remains diagnostic.

10.3 Multi-Judge Targeted Audit

The targeted multi-judge audit used MiMo as the primary context judge, DeepSeek v4 Flash as a full 192-row secondary audit filtered to the target subset, and Groq Qwen3 32B for 60 targeted rows. The strongest signal was 51/60 consensus-insufficient rows and 6 MiMo-harsh/high-disagreement rows. MiMo was not globally too harsh on this subset, but the disagreement rows should be inspected before increasing context penalties.

11 Offline Selector / Bandit-Style Baselines

The selector baselines choose among logged candidates only. They do not replace the runtime adaptive heuristic. The main policies are:

- `fixed`: choose one frequent signature; low coverage.
- `cheapest`: choose the lowest resource-cost action.
- `best_average`: choose the exact action signature with highest train mean reward.
- `family_smoothed_best_average`: back off from exact signature to retrieval-context family to context policy.
- `shrinkage_smoothed_best_average`: shrink exact/family/context means toward broader means.
- `linear_reward_model`: ridge/linear offline selector over non-leaking action and cost features.
- `smoothed_linear_selector`: add train-only aggregate means/counts to the linear selector.
- `oracle_logged`: logged-candidate upper bound, not deployable.

The six-seed answer-only sweep showed that `best_average` has the highest non-oracle reward when it covers a group, but coverage is 0.911. `shrinkage_smoothed_best_average` is the strongest full-coverage non-oracle selector so far: reward 0.602, quality 0.773, oracle gap 0.070. The learned linear selectors are functional but not yet robustly stronger than simple smoothed baselines.

Policy	Coverage	Reward	Quality	Oracle gap
cheapest	1.000	0.577	0.770	0.094
best average	0.911	0.618	0.794	0.080
shrinkage smoothed	1.000	0.602	0.773	0.070
linear reward model	1.000	0.586	0.774	0.086
smoothed linear	1.000	0.595	0.767	0.076
oracle logged	1.000	0.671	0.834	0.000

The current bottleneck is data and action coverage, not missing algorithmic complexity.

12 KV-Cache Motivation

The analytical KV-cache estimate used in the project is:

$$\text{KV bytes} \approx 2 \times L \times H_{kv} \times D_{head} \times S \times B \times \text{bytes per value}.$$

For GQA/MQA models, H_{kv} should be the number of key-value heads, not the number of attention heads. The estimates exclude model weights, activations, allocator overhead, paged-attention block overhead, and runtime workspaces.

Model	KV GiB @ 4K	KV GiB @ 32K	KV GiB @ 128K
Qwen2.5-0.5B	0.047	0.375	1.500
Qwen2.5-1.5B	0.109	0.875	3.500
Qwen2.5-3B	0.141	1.125	4.500
Qwen2.5-7B	0.219	1.750	7.000
Qwen2.5-14B	0.750	6.000	24.000

BudgetRAG is the pre-inference step: reduce or improve context before the model runs. Future MemAlign-Qwen work can compare full versus compressed context on local Qwen models, then test runtime KV retention using sink, recent, and evidence tokens.

13 Limitations

- The labels are AI feedback, not human annotations.
- The selector is offline logged-candidate evaluation, not online RL.
- Current logged action coverage is still small and retriever diversity is low.
- Current results support context-budget/action selection more strongly than retrieval-strategy selection.
- Context reward is non-default.
- Pairwise RLAIIF currently audits and calibrates scalar reward; it does not train a pairwise selector yet.
- No DPO, PPO, GRPO, online RL, or production runtime KV pruning is implemented.
- Web-search actions, if used, are live stress tests and must not be mixed with reproducible BEIR-style benchmark claims.

14 Future Work

The next steps are:

1. Inspect the 6 MiMo-harsh rows and consensus-insufficient examples.
2. Log retrieval-diversity runs with BM25, graph-BM25, and hybrid-RRF.
3. Increase query groups to reduce split variance.
4. Train context-label-aware selectors once supervision is larger.
5. Add a pairwise ranker after enough direct pair labels exist.
6. Profile local Qwen full versus compressed contexts, including prefill/decode latency and peak memory.
7. Later, test a runtime KV-pruning proof of concept that retains sink, recent, and evidence tokens.

A Appendix A: Phase 1B–1C Tables

A.1 Phase 1B Smoke Matrix

Policy / budget	Queries	Kept chars	Compression	Token savings	KV savings
char-budget 1000	10	999.7	0.2161	961.0	900.9
char-budget 2000	10	2000.0	0.4324	711.0	666.6
char-budget 4000	10	3906.0	0.8380	234.2	219.6
evidence-aware 1000	10	999.7	0.2162	961.0	900.9
evidence-aware 2000	10	2000.0	0.4324	711.0	666.6
evidence-aware 4000	10	3999.0	0.8647	234.3	219.7
legacy 1000	10	1002.0	0.2167	960.4	900.4
legacy 2000	10	2005.0	0.4336	709.6	665.2
legacy 4000	10	3914.0	0.8396	232.0	217.5
score-density 1000	10	999.7	0.2162	961.0	900.9
score-density 2000	10	2000.0	0.4324	711.0	666.6
score-density 4000	10	3906.0	0.8380	234.2	219.6

A.2 Phase 1C.2 Adaptive Profile Matrix

Profile / budget	Queries	Kept chars	Compression	Token savings	KV savings
aggressive 1000	50	999.8	0.1248	1829	1715
aggressive 2000	50	1360	0.1665	1739	1630
aggressive 4000	50	2080	0.2500	1559	1462
balanced 1000	50	2260	0.2816	1514	1419
balanced 2000	50	2840	0.3540	1369	1283
balanced 4000	50	4000	0.4991	1079	1012
conservative 1000	50	4000	0.4991	1079	1012
conservative 2000	50	4000	0.4991	1079	1012
conservative 4000	50	4000	0.4991	1079	1012
evidence-aware 1000	50	999.8	0.1248	1829	1715

evidence-aware 2000	50	2000	0.2495	1579	1480
evidence-aware 4000	50	4000	0.4991	1079	1012
legacy 1000	50	1002	0.1250	1829	1714
legacy 2000	50	2006	0.2503	1577	1479
legacy 4000	50	4014	0.5009	1075	1008

Balanced-profile diagnostics at matrix budget 1000: normalized score gap min 0.0071, mean 0.2012, max 0.5288; normalized score entropy min 0.9401, mean 0.9861, max 0.9999; score confidence min 0.00000037, mean 0.0043, max 0.0236.

B Appendix B: RLAIIF Dataset and Label Tables

B.1 Selector Smoke Build Summary

Metric	Count
RAGAS source action files	64
Completed RAGAS samples	192
Joined action-query rows	192
rlaif_actions.jsonl rows	192
rlaif_feedback.jsonl rows	192
rlaif_rewards.jsonl rows	192
rlaif_preferences.jsonl rows	578
Query groups	55
Action signatures	77
Invalid rows	0
Context-policy preferences	289
Retrieval-context preferences	289
Skipped small reward delta	694
Skipped quality guardrail failed	2

B.2 MiMo Answer Label Summary

Metric	Value
Label count	192
Valid JSON	192
Invalid JSON	0
Ambiguous labels	25
Judge errors	1
Scored labels	191
Used MiMo labels in reward	167
Fallback to original feedback	25
MiMo quality vs RAGAS correlation	0.277

Score	N	Mean	Std	Min	Max
overall_quality	191	0.823	0.307	0.000	1.000

answer_correctness	191	0.857	0.313	0.000	1.000
evidence_support	191	0.825	0.340	0.000	1.000
faithfulness	189	0.847	0.317	0.000	1.000
unsupported_claim_penalty	191	0.090	0.280	0.000	1.000

B.3 Full Context Label Validation

Shard	Rows	Clean usable
1–50	50	46
51–121	71	66
122–192	71	65
Total	192	177

Validation metric	Value
Action count	192
Merged labels	192
Missing actions	0
Unknown action ids	0
Duplicate action ids	0
Duplicate conflicts	0
Clean usable labels	177
Ambiguous labels	15
Invalid JSON labels	0
Dropped unknown chunk ids	0

B.4 Context Reward Delta Summary

Penalty	Changed	Negative	Positive	Mean all delta	Mean changed delta
0.25	140	108	32	-0.155	-0.212
0.50	140	108	32	-0.231	-0.316
1.00	140	108	32	-0.383	-0.525

Penalty	Sufficient false	Sufficient true	Missing sufficiency
0.25	64	72	4
0.50	64	72	4
1.00	64	72	4

B.5 Action Coverage Diagnostics

Level	Unique	Singleton rate	Eval group coverage
action_id	192	1.000	0.000
exact_signature	77	0.260	0.911

retrieval_context_family	16	0.000	1.000
context_policy	3	0.000	1.000
retriever	1	0.000	1.000

C Appendix C: Selector, Pairwise, and Multi-Judge Tables

C.1 Answer-Only Six-Seed Selector Sweep

Policy	Coverage	Reward	Quality	Oracle gap
fixed	0.026 ± 0.057	0.829 ± 0.000	1.000 ± 0.000	0.007
cheapest	1.000 ± 0.000	0.577 ± 0.159	0.770 ± 0.152	0.094
best average	0.911 ± 0.050	0.618 ± 0.147	0.794 ± 0.125	0.080
family smoothed	1.000 ± 0.000	0.598 ± 0.138	0.767 ± 0.122	0.074
shrinkage smoothed	1.000 ± 0.000	0.602 ± 0.117	0.773 ± 0.097	0.070
linear reward model	1.000 ± 0.000	0.586 ± 0.153	0.774 ± 0.110	0.086
smoothed linear	1.000 ± 0.000	0.595 ± 0.137	0.767 ± 0.122	0.076
oracle logged	1.000 ± 0.000	0.671 ± 0.134	0.834 ± 0.138	0.000

C.2 Context-Reward Candidate Selector Sweep

Candidate	Policy	Coverage	Reward	Quality	Oracle gap
penalty 0.25	cheapest	1.000	0.451	0.704	0.093
penalty 0.25	best average	0.911	0.478	0.701	0.085
penalty 0.25	shrinkage	1.000	0.449	0.683	0.095
penalty 0.25	smoothed linear	1.000	0.472	0.689	0.072
penalty 0.25	oracle	1.000	0.544	0.755	0.000
penalty 0.50	cheapest	1.000	0.386	0.704	0.104
penalty 0.50	best average	0.911	0.420	0.701	0.091
penalty 0.50	shrinkage	1.000	0.389	0.683	0.100
penalty 0.50	smoothed linear	1.000	0.415	0.689	0.074
penalty 0.50	oracle	1.000	0.489	0.755	0.000
penalty 1.00	cheapest	1.000	0.255	0.704	0.132
penalty 1.00	best average	0.911	0.303	0.701	0.109
penalty 1.00	shrinkage	1.000	0.273	0.689	0.114
penalty 1.00	smoothed linear	1.000	0.329	0.703	0.058
penalty 1.00	oracle	1.000	0.387	0.748	0.000

C.3 Direct Pairwise MiMo-50 Audit

Metric	Value
Pair labels	50
Valid JSON	50
Invalid JSON	0
Ambiguous labels	2
Judge/API errors	2
A wins	41

B wins	7
Agreement over non-ambiguous	0.854
Mean confidence	0.914
Small quality/support delta pairs	38
Cheaper wins when tied	35
Scalar-over-quality disagreements	5

C.4 Targeted Multi-Judge Audit

Judge	Labels	Valid	Ambiguous	Clean sufficiency
MiMo	60	60	0	57
DeepSeek	60	60	10	50
Groq	60	60	11	49

Pair	Compared	Agree	Agreement rate
MiMo vs DeepSeek	47	43	0.915
MiMo vs Groq	46	40	0.870
DeepSeek vs Groq	41	39	0.951

Audit signal	Count
Consensus insufficient cases	51
Majority vote insufficient	55
Majority vote sufficient	5
High-disagreement cases	6
MiMo-harsh cases	6

D Appendix D: KV Estimates, CLI Chronology, and Sources

D.1 Full Qwen KV-Cache Estimate Table

Model	Layers / KV heads	Seq len	KV GiB
Qwen2.5-0.5B	24 / 2	1024	0.012
Qwen2.5-0.5B	24 / 2	2048	0.023
Qwen2.5-0.5B	24 / 2	4096	0.047
Qwen2.5-0.5B	24 / 2	8192	0.094
Qwen2.5-0.5B	24 / 2	16384	0.188
Qwen2.5-0.5B	24 / 2	32768	0.375
Qwen2.5-0.5B	24 / 2	131072	1.500
Qwen2.5-1.5B	28 / 2	1024	0.027
Qwen2.5-1.5B	28 / 2	2048	0.055
Qwen2.5-1.5B	28 / 2	4096	0.109
Qwen2.5-1.5B	28 / 2	8192	0.219
Qwen2.5-1.5B	28 / 2	16384	0.438
Qwen2.5-1.5B	28 / 2	32768	0.875

Qwen2.5-1.5B	28 / 2	131072	3.500
Qwen2.5-3B	36 / 2	1024	0.035
Qwen2.5-3B	36 / 2	2048	0.070
Qwen2.5-3B	36 / 2	4096	0.141
Qwen2.5-3B	36 / 2	8192	0.281
Qwen2.5-3B	36 / 2	16384	0.562
Qwen2.5-3B	36 / 2	32768	1.125
Qwen2.5-3B	36 / 2	131072	4.500
Qwen2.5-7B	28 / 4	1024	0.055
Qwen2.5-7B	28 / 4	2048	0.109
Qwen2.5-7B	28 / 4	4096	0.219
Qwen2.5-7B	28 / 4	8192	0.438
Qwen2.5-7B	28 / 4	16384	0.875
Qwen2.5-7B	28 / 4	32768	1.750
Qwen2.5-7B	28 / 4	131072	7.000
Qwen2.5-14B	48 / 8	1024	0.188
Qwen2.5-14B	48 / 8	2048	0.375
Qwen2.5-14B	48 / 8	4096	0.750
Qwen2.5-14B	48 / 8	8192	1.500
Qwen2.5-14B	48 / 8	16384	3.000
Qwen2.5-14B	48 / 8	32768	6.000
Qwen2.5-14B	48 / 8	131072	24.000

D.2 CLI Chronology

```
rag-bench run / scripts/run_budgetrag_matrix.py
scripts/summarize_budgetrag_results.py
rag-bench rlaif-build
rag-bench rlaif-label-answers
rag-bench rlaif-label-contexts
rag-bench rlaif-reward
rag-bench rlaif-split
rag-bench rlaif-train
rag-bench rlaif-eval
rag-bench rlaif-label-pairs
scripts/summarize_rlaif_labels.py
scripts/summarize_rlaif_context_labels.py
scripts/validate_rlaif_context_labels.py
scripts/run_context_reward_ablation_pipeline.py
scripts/aggregate_rlaif_multijudge_audit.py
```

D.3 Committed Source Reports Used

- docs/reports/phase1b_smoke_results.md
- docs/reports/phase1c_adaptive_smoke_results.md
- docs/reports/phase1c1_adaptive_validation.md
- docs/reports/phase1c2_adaptive_threshold_calibration.md
- docs/reports/phase1d_rlaif_selector_smoke.md

- docs/reports/phase1d_rlaif_heldout_eval.md
- docs/reports/phase1d_rlaif_ai_judge_heldout_eval.md
- docs/reports/phase1d_rlaif_pairwise_mimo50.md
- docs/reports/phase1d_rlaif_full_context_reward_ablation.md
- docs/reports/phase1d_rlaif_multijudge_audit.md
- docs/reports/local_qwen_kv_estimates.md

E Appendix E: English Revision and Vietnamese Baseline Trace

This English report is not a literal line-by-line translation of the earlier Vietnamese monolithic report. It is a reorganized submission-oriented version. The main changes are tracked below so that future readers can compare the English report with the Vietnamese baseline from commit `ed347ee`.

Vietnamese baseline area	English report location	Change rationale
Narrative introduction and problem statement	Sections 1–2	Rewritten in English with a clearer academic problem formulation and explicit offline-evaluation boundary.
Related work paragraphs	Section 3	Kept the same citation clusters, but separated RAG/RAGAS, compression, KV-cache, graph retrieval, RLAIIF, and contextual bandits.
Long monolithic implementation description	Sections 4–7	Split into system overview, Phase 1A–1C, generation feedback, and RLAIIF pipeline.
Main result paragraphs	Section 8	Consolidated the strongest numbers near the front: context labels, reward ablation, and multi-judge audit.
Selector discussion	Section 9 and Appendix C	Kept the conclusion that learned selectors are data-limited; moved detailed sweep tables to appendix.
KV-cache discussion	Section 10 and Appendix D	Kept formula and Qwen estimates; added full appendix table instead of only the compact table.
Sparse original appendix	Appendices A–D	Expanded with Phase 1B/1C matrices, RLAIIF build stats, answer/context labels, reward deltas, selector sweeps, pairwise audit, multi-judge audit, and full KV estimates.

Claim boundaries	Sections 11–12 and Appendix E	Preserved and made explicit: no human labels, no online RL, no DPO/PPO/GRPO, no production KV pruning.
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Tracking policy for future edits:

- Keep `pdf/main.tex` as a short driver only.
- Put English content under `pdf/sections/en/`.
- Add new report data to appendices first, then summarize only the strongest result in the main body.
- If a Vietnamese version is needed later, create `pdf/sections/vi/` rather than mixing languages in the same section files.
- Keep raw experiment files under `ignored benchmark_results/`; commit only curated tables and reports.

References

- [1] Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Andy Jones, Anna Chen, Anna Goldie, Azalia Mirhoseini, Cameron McKinnon, Carol Chen, Catherine Olsson, Chris Olah, Danny Hernandez, Dawn Drain, Deep Ganguli, Dustin Li, Eli Tran-Johnson, Ethan Perez, Jamie Kerr, Jared Mueller, Jeffrey Ladish, Joshua Landau, Kamal Ndousse, Kamile Lukosuite, Liane Lovitt, Michael Sellitto, Nelson Elhage, Nicholas Schiefer, Nova DasSarma Mercado, Robert Lasenby, Robin Larson, Sam Ringer, Scott Johnston, Shauna Kravec, Sheer Showk, Stanislav Fort, Tamera Lanham, Timothy Telleen-Lawton, Tom Conerly, Tom Henighan, Tristan Hume, Samuel R. Bowman, Zac Hatfield-Dodds, Ben Mann, Dario Amodei, Nicholas Joseph, Sam McCandlish, Tom Brown, and Jared Kaplan. Constitutional ai: Harmlessness from ai feedback. *arXiv preprint arXiv:2212.08073*, 2022. <https://arxiv.org/abs/2212.08073>.
- [2] Zefan Cai, Yichi Zhang, Bofei Gao, Yuliang Liu, Tianyi Liu, Keming Lu, Wayne Xiong, Yue Dong, Baobao Chang, Junjie Hu, and Wen Xiao. Pyramidkv: Dynamic kv cache compression based on pyramidal information funneling. *arXiv preprint arXiv:2406.02069*, 2024. <https://arxiv.org/abs/2406.02069>.
- [3] Gordon V. Cormack, Charles L. A. Clarke, and Stefan Buettcher. Reciprocal rank fusion outperforms condorcet and individual rank learning methods. In *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 758–759, 2009. <https://doi.org/10.1145/1571941.1572114>.
- [4] Darren Edge, Ha Trinh, Newman Cheng, Joshua Bradley, Alex Chao, Apurva Mody, Steven Truitt, and Jonathan Larson. From local to global: A graph rag approach to query-focused summarization. *arXiv preprint arXiv:2404.16130*, 2024. <https://arxiv.org/abs/2404.16130>.
- [5] Shahul Es, Jithin James, Luis Espinosa-Anke, and Steven Schockaert. Ragas: Automated evaluation of retrieval augmented generation. *arXiv preprint arXiv:2309.15217*, 2023. <https://arxiv.org/abs/2309.15217>.
- [6] Huiqiang Jiang, Qian Wu, Chin-Yew Lin, Yuqing Yang, and Lili Qiu. Llmllingua: Compressing prompts for accelerated inference of large language models. *arXiv preprint arXiv:2310.05736*, 2023. <https://arxiv.org/abs/2310.05736>.

- [7] Huiqiang Jiang, Qian Wu, Xufang Luo, Dongsheng Li, Chin-Yew Lin, Yuqing Yang, and Lili Qiu. Longllmlingua: Accelerating and enhancing llms in long context scenarios via prompt compression. *arXiv preprint arXiv:2310.06839*, 2023. <https://arxiv.org/abs/2310.06839>.
- [8] Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Kuttler, Mike Lewis, Wen-tau Yih, Tim Rocktaschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive nlp tasks. *arXiv preprint arXiv:2005.11401*, 2020. <https://arxiv.org/abs/2005.11401>.
- [9] Lihong Li, Wei Chu, John Langford, and Robert E. Schapire. A contextual-bandit approach to personalized news article recommendation. *arXiv preprint arXiv:1003.0146*, 2010. <https://arxiv.org/abs/1003.0146>.
- [10] Yucheng Li, Bo Dong, Frank Guerin, and Chenghua Lin. Compressing context to enhance inference efficiency of large language models. *arXiv preprint arXiv:2310.06201*, 2023. <https://arxiv.org/abs/2310.06201>.
- [11] Yuhong Li, Yingbing Huang, Bowen Yang, Bharat Venkitesh, Andrea Locatelli, Hanchen Ye, Tian Cai, Patrick Lewis, and Deming Chen. Snapkv: Llm knows what you are looking for before generation. *arXiv preprint arXiv:2404.14469*, 2024. <https://arxiv.org/abs/2404.14469>.
- [12] Zirui Liu, Jiayi Yuan, Hongye Jin, Shaochen Zhong, Zhaozhuo Xu, Vladimir Braverman, Beidi Chen, and Xia Hu. Kivi: A tuning-free asymmetric 2bit quantization for kv cache. *arXiv preprint arXiv:2402.02750*, 2024. <https://arxiv.org/abs/2402.02750>.
- [13] Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. *arXiv preprint arXiv:2305.18290*, 2023. <https://arxiv.org/abs/2305.18290>.
- [14] Nandan Thakur, Nils Reimers, Andreas Ruckle, Abhishek Srivastava, and Iryna Gurevych. Beir: A heterogeneous benchmark for zero-shot evaluation of information retrieval models. *arXiv preprint arXiv:2104.08663*, 2021. <https://arxiv.org/abs/2104.08663>.
- [15] Guangxuan Xiao, Yuandong Tian, Beidi Chen, Song Han, and Mike Lewis. Efficient streaming language models with attention sinks. *arXiv preprint arXiv:2309.17453*, 2023. <https://arxiv.org/abs/2309.17453>.
- [16] Zhenyu Zhang, Ying Sheng, Tian Zhou, Tian Chen, Lianmin Zheng, Ruisi Cai, Zhao Song, Yuandong Tian, Christopher Re, Clark Barrett, Zhangyang Wang, and Beidi Chen. H2o: Heavy-hitter oracle for efficient generative inference of large language models. *arXiv preprint arXiv:2306.14048*, 2023. <https://arxiv.org/abs/2306.14048>.